PharmAchieve

CHRONIC FATIGUE SYNDROME

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

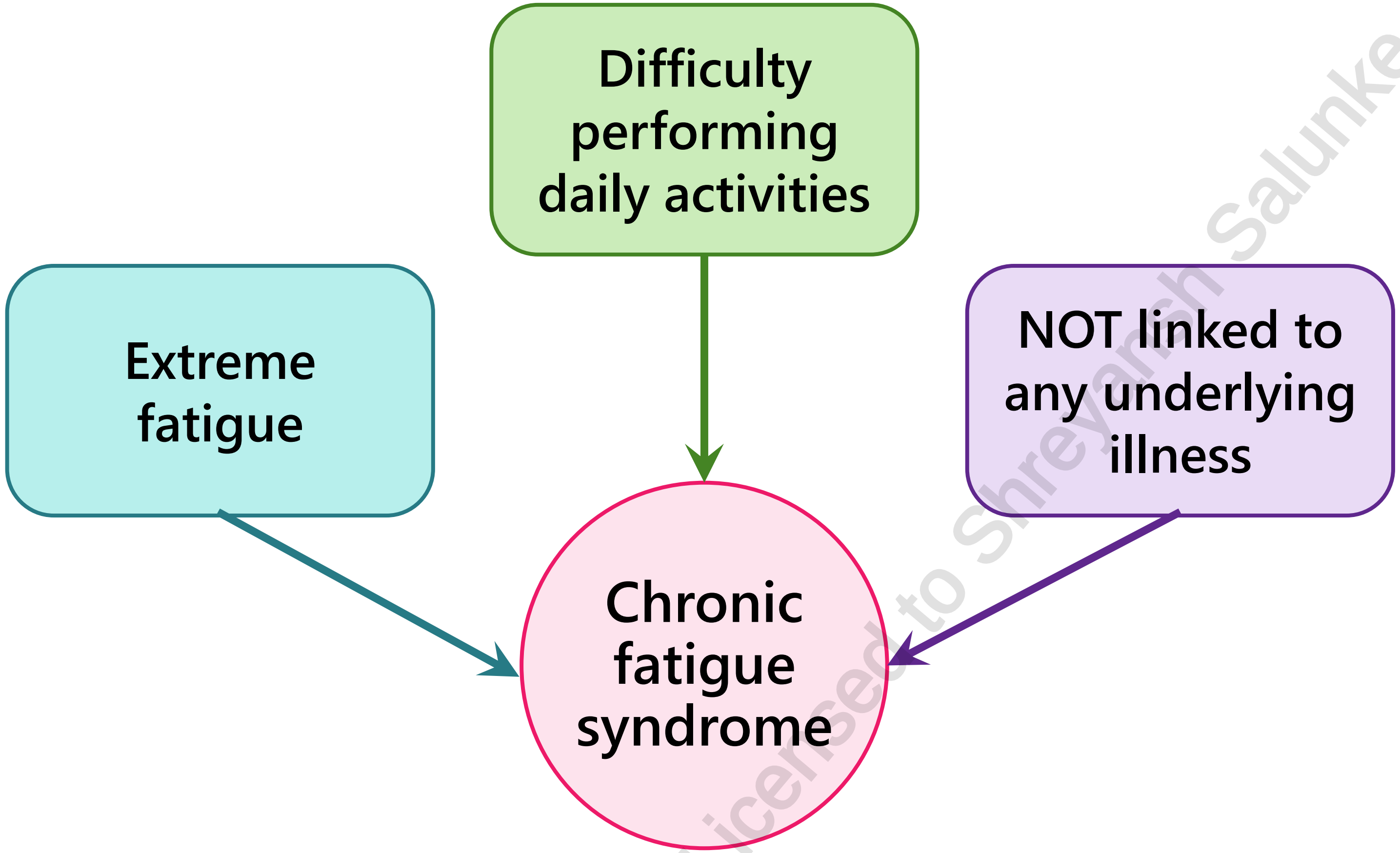
Communication
& Collaboration

Leadership &
Stewardship

Professionalism

WHAT IS CHRONIC FATIGUE SYNDROME?

Also referred to as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)



- Resting does NOT improve symptoms
- Physical or mental activities may worsen symptoms
- The direct cause of the disorder is unknown however some believe that the cause is multifactorial
- Diagnosis is based on “ruling out” other conditions with similar symptoms

CLINICAL PRESENTATION AND RISK FACTORS

Clinical presentation

- Severe chronic fatigue occurring for long periods of time
- Post-exertional malaise (*cardinal symptom*) – general discomfort and weakness following physical or mental exertion with slow recovery (>24 hours)
- Muscle and/or joint pain
- Sleep dysfunction (unrefreshing sleep or disturbances in sleep quantity or rhythm)
- Neurological/cognitive dysfunction (memory and concentration problems)
- Autonomic/immune dysfunction: orthostatic intolerance (light-headedness upon standing), gastrointestinal abnormalities, heart rate abnormalities, hypersensitivity to light and sound, headache

Risk factors

- Ethnic minorities
- Women > men
- Age 40-60 years
- Immune response to viral infections (e.g. Epstein-Barr virus, human herpes virus-6)
- Lower socioeconomic class, stress/trauma, possible genetic link

GOALS OF THERAPY

- ✓ Build trust and rapport with patients
- ✓ Discuss symptoms that patients may want to treat first and create an individualized plan
- ✓ Help patients improve quality of life and cope with the disorder through education and practical strategies

ASSESSMENT

New onset, unexplained persistent fatigue significantly affecting daily activities

→

Perform history, physical, neurologic, psychiatric and mental status exams; perform routine lab tests and any additional tests required

→

If lab and exam results are normal, assess for other possible causes (e.g. medical conditions, medications)

If lab and exam results are not normal, assess symptoms and treat

NON-PHARMACOLOGICAL THERAPY

- Chronic fatigue syndrome does not currently have a cure
- Therapy focuses on symptomatic treatment: patients are given an individualized integrative therapy regimen consisting of medications, supplements and non-pharmacological measures
- Encourage a healthy, balanced diet with adequate protein, fresh vegetables and fruit

Clinical Presentation	Non-Pharmacological Measures
Post-exertional malaise	• Activity pacing: spacing out activities throughout the day (e.g. spoon theory)
Sleep dysfunction	Sleep hygiene – developing good sleeping habits. This includes: • Keep a regular sleeping/waking schedule with a consistent waking time 7 days/week • Avoid sleeping in, limit daytime napping if possible (although sometimes needed) • Avoid caffeine, nicotine, alcohol and other recreational drugs • Use the bedroom for sleep and intimacy only
Chronic fatigue	• Cognitive behavioural therapy – offered as a supportive psychological tool to assist in symptom management
Orthostatic intolerance	• Assess the symptom (e.g. does the patient feel light-headed when they stand up from a seated position?)

PHARMACOLOGICAL THERAPY

- Mixed evidence for use of complementary and alternative medicines:
  - Few negative outcomes from these supplements
  - Supplementation may benefit some patients
  - Include L-carnitine, oral nicotinamide adenine dinucleotide, magnesium, D-ribose, omega-3 fatty acids, vitamin D

Poor evidence for all listed options

- If sleep hygiene is not helpful + no diagnosis for a specific sleep disorder, the following can be considered:
  - Short trial of sedating antihistamines (such as doxylamine) OR
  - Prescribed hypnotic at the lowest dose
- Rituximab showed some efficacy in a smaller study but due to its side effect profile and lack of robust data it is not recommended to be used as pharmacotherapy

Stimulants (e.g. modafinil and methylphenidate), and tricyclic antidepressants are not recommended

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	**Degree of Change**	**Timeframe**
Fatigue, autonomic disturbances and psychosocial dysfunction	Improvement	Ongoing

Safety endpoints		
Parameter	**Degree of Change**	**Timeframe**
Side effects from complementary alternative therapy (various)	Prevent	Duration of therapy

TAKEAWAY

- Chronic fatigue syndrome is not linked to an underlying cause and may present with difficulty in performing daily activities and severe chronic fatigue for long periods of time
- Diagnosis is based on ruling out other conditions with similar symptoms
- There is currently no cure
- Symptomatic management:
  - Non-pharmacologic options include activity pacing, sleep hygiene, and cognitive behavioural therapy
  - Pharmacologic options include a short trial of antihistamines (e.g. doxylamine) or a prescribed hypnotic at the lowest dose
- Complementary and alternative medicines have mixed evidence. Examples include: L-carnitine, oral nicotinamide adenine dinucleotide, magnesium, D-ribose, omega-3 fatty acids, vitamin D
- Not recommended: rituximab, stimulants, tricyclic antidepressants

CASE STUDY

PL is a 36-year-old Inuit female who presents to the pharmacy complaining of persistent fatigue for the past 8 months. She explains that despite having adequate rest, she experiences difficulties in concentrating at work and performing activities during her daily routine at home, and this contributes to her increased levels of stress. PL explains that she follows a healthy, balanced diet and she exercises once a week for 30 minutes. Following her weekly exercise routine, she experiences extreme malaise for the next 48 hours. Her symptoms began approximately 8 months ago but they have gradually worsened over time. She is not currently pregnant or breastfeeding, and she has no known allergies to any medications. Her past medical history is insignificant. PL takes a multivitamin supplement PO daily. She denies any alcohol, smoking, or recreational drug use.

Her physician has been regularly monitoring her laboratory results since her symptoms began and her most recent laboratory results returned today revealing a serum ferritin level of 330 pmol/L (reference: 54 to 690 pmol/L), a hemoglobin level of 150 g/L (reference: 120 to 160 g/L), a thyroid-stimulating hormone (TSH) level of 3 mIU/L (reference 0.5 to 4.0 mIU/L), and a free thyroxine (fT4) level of 18.5 pmol/L (reference: 10.3 to 23.2 pmol/L). PL's physician rules out all other possible causes before diagnosing PL with chronic fatigue syndrome.

CASE STUDY - QUESTIONS

1. Which of the following is NOT a risk factor that PL has for the development of chronic fatigue syndrome?
   a) Female sex
   b) Ethnicity
   c) Age
   d) Stress
2. Which of the following statements is FALSE regarding chronic fatigue syndrome?
   a) There is no cure
   b) An individualized plan should be developed to manage chronic fatigue syndrome
   c) Omega-3 fatty acids may provide some benefit
   d) Dietary modifications have no place in therapy
3. Which of the following options is LEAST appropriate to recommend to PL to manage her symptoms?
   a) Cognitive behavioural therapy
   b) Activity pacing
   c) Sleep hygiene measures
   d) A short trial of doxylamine
4. PL asks about the pharmacologic therapies that may be available to her. What is the most appropriate advice to tell PL?
   a) Pharmacotherapy may not provide benefit at this time
   b) Recommend methylphenidate
   c) Recommend nortriptyline
   d) Recommend rituximab

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CHANGE LOG

March 2025

- Slide 4: Updated clinical presentation
- Slide 7: Added a table for non-pharmacological therapy
- Slide 10: Added references

July 2025

- Formatting, grammar, spelling changes throughout
- Slide 4: Added definition of sleep dysfunction
- Slide 10: Takeaway slide added
- Slides 11, 12: Case study and questions added
- Slide 13: Updated references

November 2025

- Slide 3: Added another name for the condition
- Slide 9: Updated efficacy endpoints

January 2026

- Slides 7, 10: Removed exercise therapy
- Slide 12: Modified answer options for question 3

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 7: Minor updates